

Perceptually Uniform Hue Adjustment: Hue Distortion Cage

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Figure 1: Shifting red hues $+180^\circ$ of an image (left) using our method (centre) compared to common software in Lab color mode (right). The plots show a gradient of h in OKLch, with the lower half showing the adjustment applied using each method.

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1 Introduction

Hue Adjustments are a common category of tool available in many photo editing software packages; however, these tools tend to yield unexpected colors when hues are shifted to their extrema ($\pm 180^\circ$). In our assessment, there are two primary deficiencies of such tools: 1) the weighting and shifting operations are not performed in a perceptually uniform manner, causing variance in luminance and chrominance across hue gradients, and 2) the shifting operation is performed in a polar coordinate space, introducing hue cycling between shifted hue angles.

Many software packages mask these deficiencies by constraining the hue rotation angle to $\pm 30^\circ$, which improves predictability at the cost of the functionality of the tool.

We present a method for making perceptually linear hue adjustments, called Hue Distortion Cage (HDC). Leveraging the OK-Lab [Ottooson 2021] color space, our approach involves perceptually uniform interpolation of affected colors in the L,a,b opponent model, rather than simply shifting the hue in the polar L,c,h model. As

our method operates in a perceptually-aware coordinate space, hue shifts do not yield any perceived variance in luminance or chromaticity. Additionally, as hue selections are made in a perceptually-aware manner, the weighting of adjustments is more predictable compared to other approaches.

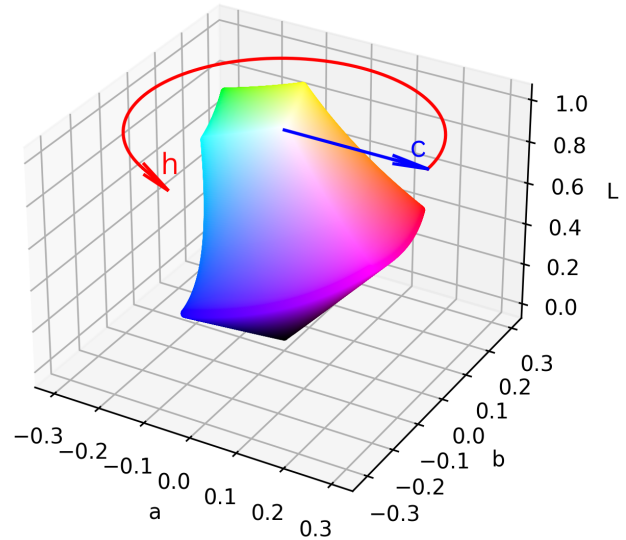


Figure 2: sRGB gamut plotted in the OKLab color space. Our method uses the h term of the OKLch model to make weighted pixel selections, and interpolates colors in the OK-Lab space.

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2 Our Method

We first define a fixed number of segments N at equidistant hue intervals of $\frac{2\pi}{N}$ in the OKLch color space, which correspond to the shiftable hues (e.g., reds, yellows, greens, ...).

For a given sRGB sample, we determine its hue h in the OKLch color space, and then identify the two nearest hue segments S_a and S_b to h . Because the hues are arranged in a circular manner, we wrap the segment indices around, such that if S_a is the last segment, S_b is the first segment.

Let $H = \{h_1, h_2, \dots, h_N\}$ be the set of segment hues at equidistant intervals. The segments S_a and S_b are defined as:

$$\begin{aligned} S_a &= \max\{h_i \in H \mid h_i \leq h\} \\ S_b &= \min\{h_i \in H \mid h_i \geq h\} \\ \text{if } S_b &= \emptyset, \text{ then } S_b = h_1 \end{aligned} \quad (1)$$

The distance α is then given by the fractional position of h between S_a and S_b . Since the hues are circular, if $S_a > S_b$, we add 2π to α :

$$\alpha = \begin{cases} \frac{h - S_a}{S_b - S_a} & \text{if } S_a \leq S_b \\ \frac{h - S_a + 2\pi}{S_b - S_a + 2\pi} & \text{if } S_a > S_b \end{cases} \quad (2)$$

Each hue segment is associated with a user-adjustable "Control Point." A Control Point is a tuple $(\Delta L, \Delta c, h')$, where ΔL and Δc are offset amounts for luma and chroma, respectively, and h' is the hue value to be set for that segment.

Using the Control Points, we derive the hue shifted colors A_{hue} and B_{hue} , which are the original sample color but with their hues set to the S_a and S_b Control Point values.

$$\begin{aligned} A_{hue} &= (L_a, c_a, h'_a) \\ B_{hue} &= (L_b, c_b, h'_b) \end{aligned} \quad (3)$$

A key distinction of our approach is the method of interpolating between the adjusted Control Points. Instead of rotating in OKLch polar space, we perform linear interpolation in the OKLab space. This method ensures that the colors transition through the OKLab volume in a uniform manner, avoiding cyclic hues in gradients between colors.

$$X_{lab} = (1 - \alpha)A_{lab} + \alpha B_{lab} \quad (4)$$

Where $A_{lab} = (L_{Ahue}, C_{Ahue} \cos h_{Ahue}, C_{Ahue} \sin h_{Ahue})$ and $B_{lab} = (L_{Bhue}, C_{Bhue} \cos h_{Bhue}, C_{Bhue} \sin h_{Bhue})$, with L as the lightness, C as the chroma, and h as the hue of the shifted colors in OKLch space. The parameter α is the interpolation factor.

After performing interpolation in OKLab space, the interpolated color X_{lab} represents the hue-shifted sample color. We now interpolate the control points A and B to derive the corresponding adjustments for luma (L) and chroma (c) using the previously calculated interpolation fraction α . This yields the interpolated control point X .

X therefore holds the resultant $(\Delta L, \Delta c)$ which can be used as weighting values to apply brightness and saturation adjustments, for example.

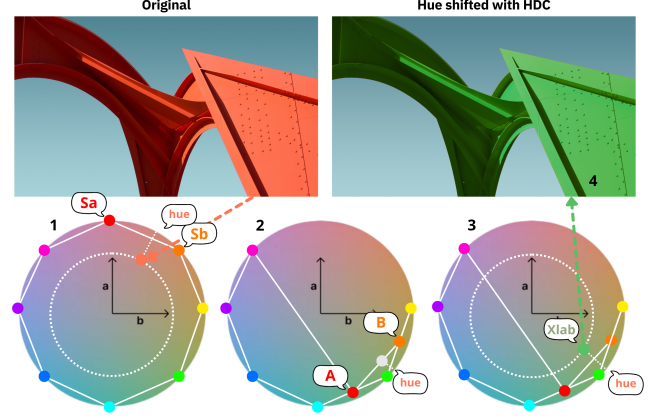


Figure 3: Annotated process of applying HDC to an image. (1) The OKLch hue of the sample and its neighbouring hue sectors S_a and S_b are found. (2) In this example, both A and B control points are rotated. (3) X_{lab} is found by interpolating between A and B using α , and its c and L components are inherited from the sample. (4) The X_{lab} color is converted back to sRGB. Photo by Pierre Châtel-Innocenti on Unsplash



Figure 4: Nonlinear brightness adjustment applied (left) using the brightness influence mask, derived from ΔL_X (right) where the ΔL term for the Control Point at $h = 0$ (red) has been set at -1.0 .

References

Björn Ottosson. 2021. *A perceptual color space for image processing*. <https://bottosson.github.io/posts/oklab/> Accessed: 2024-07-09.